PROBLEM-1

$(X, Y) \in \underbrace{S \times \{+1, -1\}}_{\text{Some distribution } D}$ → $P(Y|X)$ has to be Bernoulli.

$$\eta(x) := E[Y \mid X = x]$$

$$= \sum_{y=\{+1,-1\}} y P(Y = y \mid X = x)$$

$$= P(Y=1 \mid X = x) - P(Y=-1 \mid X = x)$$

$$= p_x - (1 - p_x) = 2p_x - 1$$

where p_x = probability of flipping +1 at $X = x$

BOP — Bayes optimal predictor.

$$g_*(x) = \text{Sign}(\eta(x)) = \text{Sign}(2p_x - 1)$$

$$\therefore g_*(x) = \begin{cases} +1 & \text{if } p_x \geq 1/2 \\ -1 & \text{otherwise.} \end{cases}$$

* Interpretation: At $X = x$, if it is more likely to toss a coin at get heads ($Y = +1$), then BOP always predicts Heads.

* g_* is a function from $X \to Y$ and can only output a single value for a given $X = x$.

* To be correct more often than not, g_* tries to predict always the same outcome towards which the coin is biased (P_x) at $X = x$.

Risk

Let $g : X \to \{+1, -1\}$ be any predictor.

Let $\ell : \{+1, -1\} \times \{-1, +1\} \to \mathbb{R}_{\geq 0}$ (loss function)

The risk of any predictor g is given by

$$L(g) = \mathbb{E}_{(X,Y)\sim D}[\ell(Y, g(X))]$$

$$= E_{X\sim \pi}[E_{Y\sim P(\cdot|X)}[\ell(Y, g(X))]]$$

marginal

$$\pi(x) = \int D(x,y)\, dy$$

$$\| $$

$$\sum_{y=\{+1,-1\}} D(x,y)$$

Clearly, the risk depends on the loss function 'ℓ'.

In Classification, it is taken to be the 0-1 loss.

i.e. $\ell(Y, g(X)) = \mathbb{1}_{\{Y \neq g(X)\}}$

Bayes Risk: The Risk of the Bayes Optimal Predictor

$\therefore$ $L(g_*) = E[\ell(Y, g_*(X)] := L_*$

We want this risk under the 0-1 loss.

$$L_* = \text{Bayes Risk} = E_{(X,Y)\sim D}\left[\ell(Y, g_*(X))\right]$$

$$L_* = \mathbf{E}_{X\sim\Pi}\, E_{Y\sim P(\cdot|X)}\left[\ell(Y, g_*(X))\right]$$

Recall that $P(Y|X)$ is Bernoulli, $\text{Ber}(p_x)$ where p_x is the probability of flipping $Y=+1$ at $X=x$.

$$\therefore \text{Bayes Risk} = E_{X\sim\Pi}\, E_{Y\sim P(\cdot|X)}\left[\ell(Y, g_*(X))\right]$$

Recall

$$g_*(x) = \text{sign}(2p_x - 1)$$

$$\text{So } g_*(x) = \begin{cases} +1 & \text{if } p(x) := p_x \geq 1/2 \\ -1 & \text{otherwise.} \end{cases}$$

$$\therefore \ell(Y, g_*(x)) = \mathbb{1}\{Y \neq g_*(x)\} \times 1$$

$$= \mathbb{1}\{Y \neq g_*(x)\} \times \left(\mathbb{1}\{p_x \geq 1/2\} + \mathbb{1}\{p_x < 1/2\}\right)$$

$$= \mathbb{1}\{Y \neq g_*(x)\} \times \mathbb{1}\{p_x \geq 1/2\} + \mathbb{1}\{Y \neq \overset{-1}{\uparrow} g_*(x)\}$$

$$\downarrow$$
$$+1$$

$$\mathbb{1}\{p_x < 1/2\}$$

$$\Rightarrow \ell(Y, g_*(x)) = 1\{Y \neq +1\}\, 1\{p_x \geq 1/2\} + 1\{Y \neq -1\}\, 1\{p_x < 1/2\}$$

Now,

$$E_{Y \sim P(\cdot|x)}\left[\ell(Y, g_*(x))\right]$$

$$= E_{Y \sim P(\cdot|x)}\left[\, 1\{Y \neq +1\}\, 1\{p_x \geq 1/2\} + 1\{Y \neq -1\}\, 1\{p_x < 1/2\}\,\right]$$

expectation w.r.t y when X is held fixed at x.

For a given x, $p_x = p(x)$ is constant ⇒ So are any function of p_x ⇒ these are constant wrt Y too ⇒ can be pulled out of expectation.

$$= 1\{p_x \geq 1/2\}\, E_{Y \sim P(\cdot|x)}\left[\, 1\{Y \neq +1\}\,\right] + 1\{p_x < 1/2\}\, E_{Y \sim P(\cdot|x)}\left[\, 1\{Y \neq -1\}\,\right]$$

(expectation of Indicator) $P(Y \neq +1 \mid X = x)$

(expectation of Indicator) $P(Y = -1 \mid X = x)$

$$= 1\{p_x \geq 1/2\}\, P(Y \neq +1 \mid X = x) + 1\{p_x < 1/2\}\, P(Y \neq -1 \mid X = x)$$

Recall that $P(Y=+1 \mid X=x) = p_x$.

$\therefore P(Y \neq +1 \mid X=x) = 1 - p_x$.

Similarly $P(Y=-1 \mid X=x) = 1 - p_x$

& $P(Y \neq -1 \mid X=x) = 1 - (1-p_x) = p_x$.

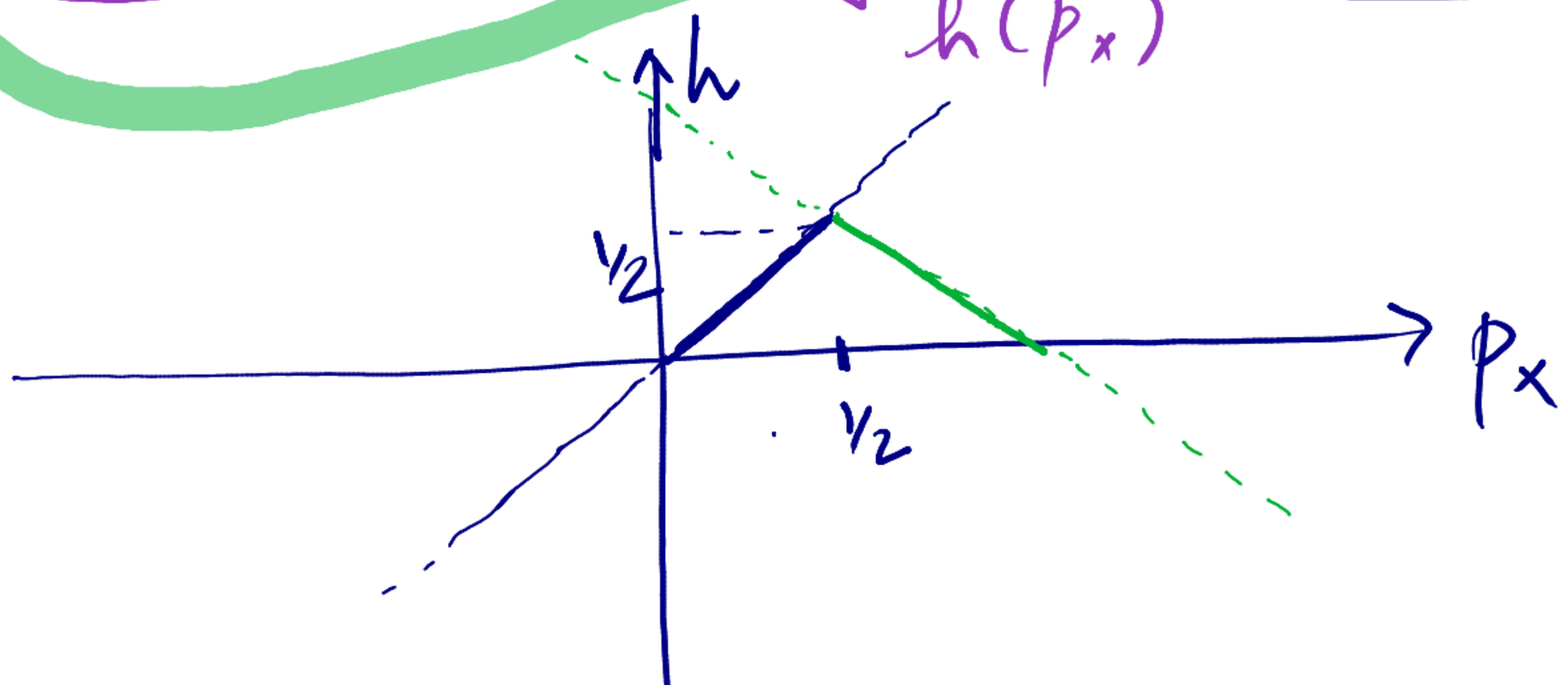
$\therefore E_{Y \sim P(\cdot|x)}[\ell(Y, g_*(x))]$

$= 1_{\{p_x \geq 1/2\}} P(Y \neq +1 \mid X=x)$

$+ 1_{\{p_x < 1/2\}} P(Y \neq -1 \mid X=x)$

$= 1_{\{p_x \geq 1/2\}} (1-p_x) + 1_{\{p_x < 1/2\}} p_x$

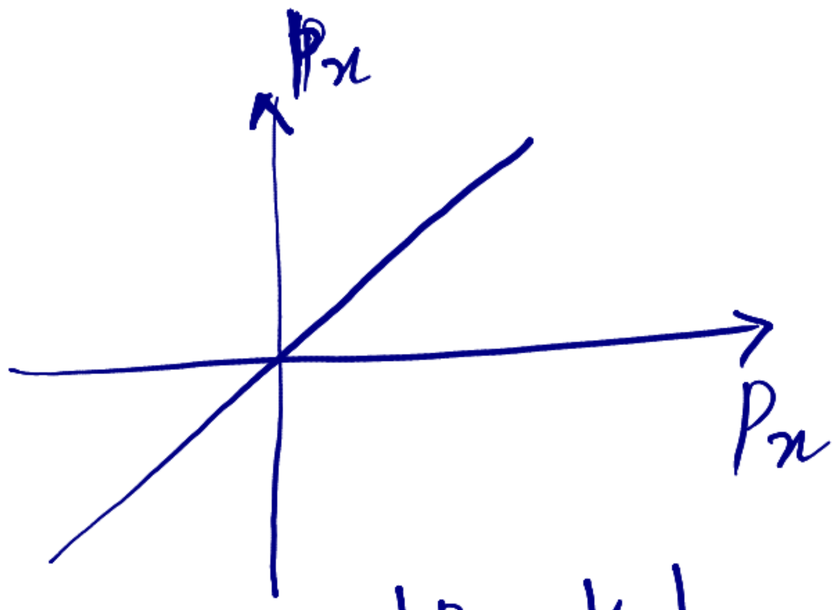
$h(p_x)$

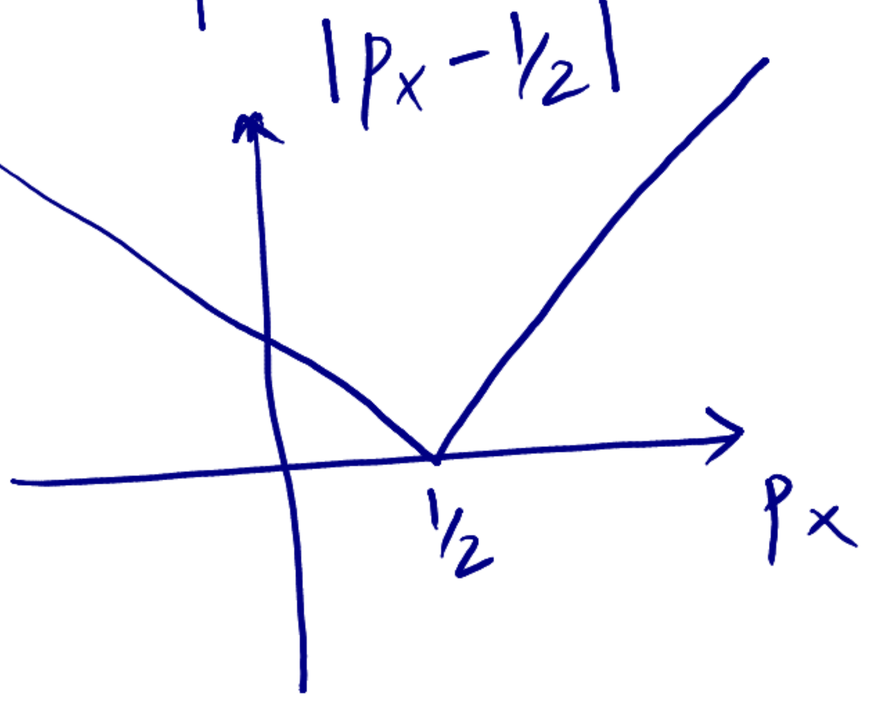


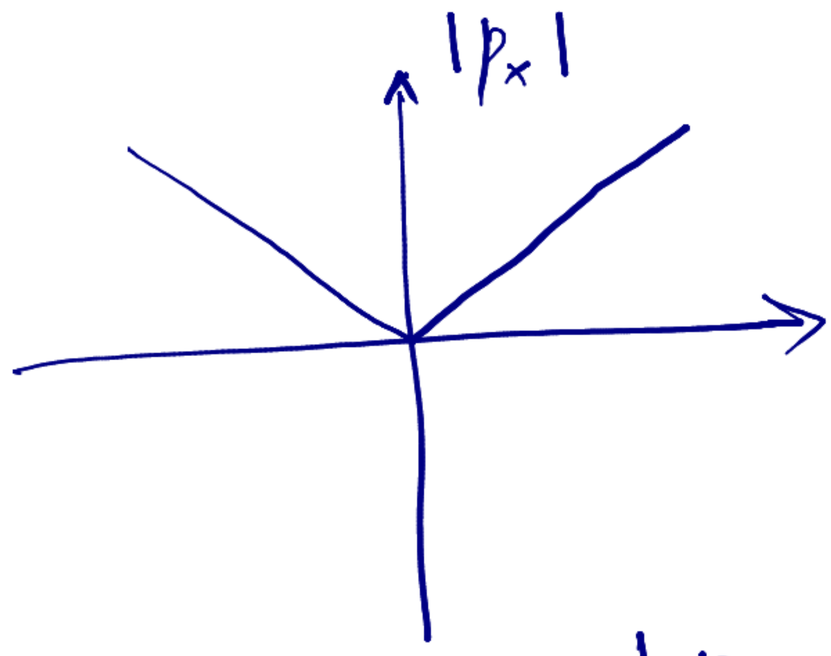
This is same as

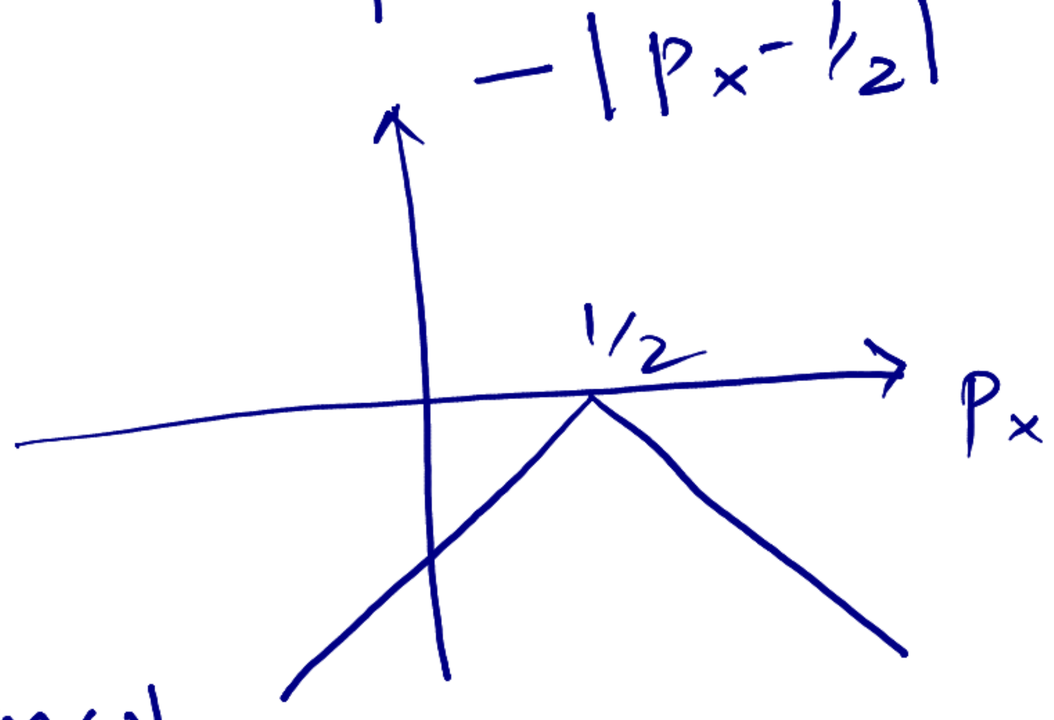
$$h(p_x) = \min\{p_x, 1-p_x\}$$

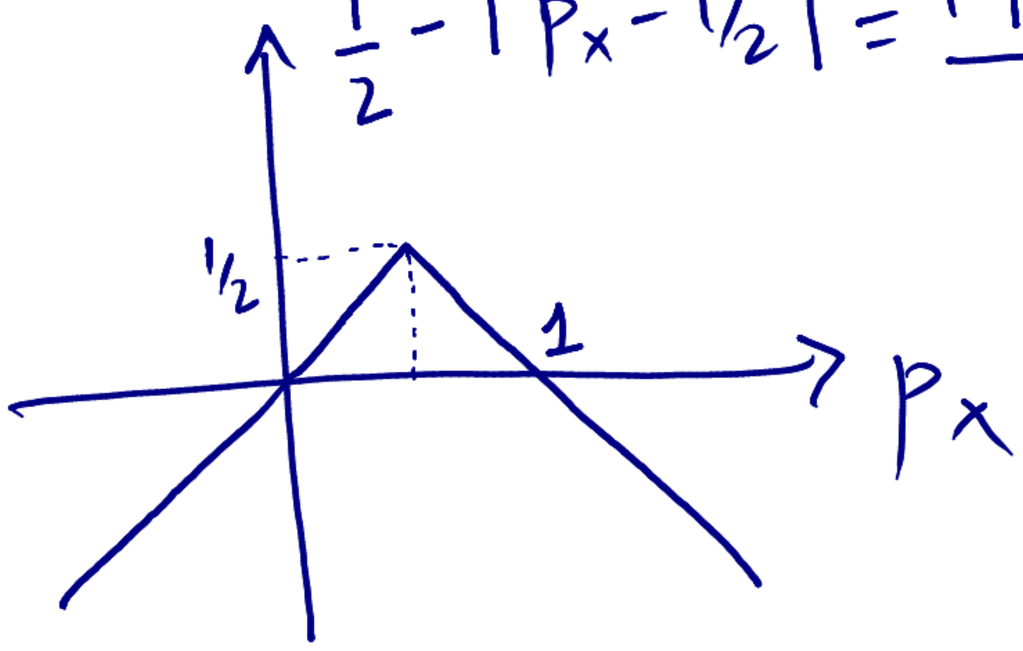
and also $h(p_x) = 1/2 - |p_x - 1/2|$











$$= \frac{1-|\eta(x)|}{2}$$

Recall $\eta(x) = 2p_x - 1$

$$\Rightarrow p_x - \frac{1}{2} = \frac{\eta(x)}{2}$$

$$\Rightarrow |p_x - 1/2| = \frac{|\eta(x)|}{2}$$

$$\Rightarrow \frac{1}{2} - |p_x - 1/2| = \frac{1}{2} - \frac{|\eta(x)|}{2}$$

$$= \frac{1-|\eta(x)|}{2}$$

Hence $E_{Y\sim P(\cdot|x)}[\ell(Y, g_*(x))]$

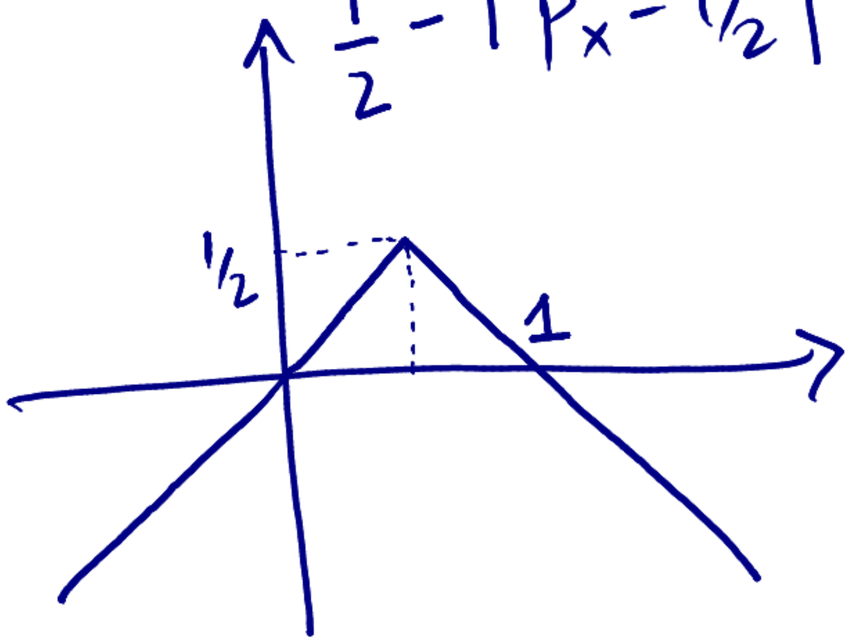
$$= \frac{1-|\eta(x)|}{2}$$

Now; $L_* =$ Bayes Risk

$$= E_{X\sim\pi}[E_{Y\sim P(\cdot|x)}[\ell(Y, g_*(x))]]$$

$$= E_{x\sim\pi}\left[\frac{1-|\eta(x)|}{2}\right]$$

(1) Recall the figure below.



$\forall\ p_x \in [0, 1]$,

$$\frac{1-|\eta(x)|}{2} \le 1/2$$

$$\Rightarrow E_{x\sim\pi}\left[\frac{1-|\eta(x)|}{2}\right] \le E_{x\sim\pi}[1/2]$$

$$\Rightarrow \underbrace{E\left[\frac{1-|\eta(x)|}{2}\right]}_{} \le 1/2 \qquad 1/2$$

$$\Rightarrow L_* \le 1/2$$

Since $L^* \leq 1/2$, the statement in the question that Bayes Risk L_* can take any value between $[0, 1]$ is false since it cannot take values above $1/2$.

② Excess Risk :-

The excess risk of a predictor $g(X)$ is given by how much more risk it carries over the Bayes Risk L_*.

$$\text{Excess Risk} = \mathcal{E}(g) = L(g) - L_*$$

where $L(g) = E_{(X,Y)\sim D}[\ell(Y, g(X))]$

and $L_* = \min_h E_{(X,Y)\sim D}[\ell(Y, h(X))]$

Obviously, $g_* = \operatorname{argmin}_h E_{(X,Y)\sim D}[\ell(Y, h(X))]$

$$\mathcal{E}(g) = L(g) - L_* = L(g) - L(g_*)$$

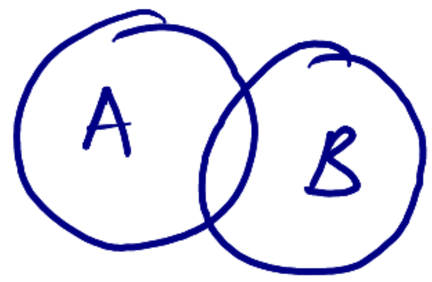
$$= E_{(X,Y)\sim D}[\ell(Y,g)] - E_{(X,Y)\sim D}[\ell(Y,g_*)]$$

$$= E_{(X,Y)\sim D}[\ell(Y,g) - \ell(Y,g_*)]$$

$$= E_{(X,Y)\sim D}[\mathbb{1}_{\{Y \neq g(X)\}} - \mathbb{1}_{\{Y \neq g_*(X)\}}]$$

Recall that if A & B are sets,

$$\mathbb{1}_A - \mathbb{1}_B = \mathbb{1}_{A\setminus B} - \mathbb{1}_{B\setminus A}$$



$A \equiv Y \neq g(X)$ $\quad B \equiv Y \neq g_*(X)$

$\bar{A} \equiv Y = g(X)$ $\quad \bar{B} \equiv Y = g_*(X)$

$$A\setminus B \equiv A \cap \bar{B} \equiv \underbrace{\{Y \neq g(X)\} \cap \{Y = g_*(X)\}}$$

$$\equiv \{g(X) \neq Y = g_*(X)\}$$

Similarly $B\setminus A \equiv B \cap \bar{A} \equiv \underbrace{\{Y \neq g_*(X)\} \cap \{Y = g(X)\}}$

$$\equiv \{g(X) = Y \neq g_*(X)\}$$

Let $C = \{g(X) \neq g_*(X)\}$

$\therefore$ Then $A \setminus B = A \cap \bar{B} = \{ Y \neq g(x) \} \cap \{ Y = g_*(x) \}$

$= \{ g_*(x) \neq g(x) \} \cap \{ Y = g_*(x) \}$

Also $B \setminus A = B \cap \bar{A}$

$= \{ Y \neq g_*(x) \} \cap \{ Y = g(x) \}$

$= \{ g(x) \neq g_*(x) \} \cap \{ Y = g(x) \}$

$\therefore \mathbb{I}_A - \mathbb{I}_B = \mathbb{I}_{A \setminus B} - \mathbb{I}_{B \setminus A}$

$= \mathbb{I}_{\{ g_*(x) \neq g(x) \} \cap \{ Y = g_*(x) \}}$

$- \mathbb{I}_{\{ g(x) \neq g_*(x) \} \cap \{ Y = g(x) \}}$

Since $\mathbb{I}_{P \cap Q} = \mathbb{I}_P \cdot \mathbb{I}_Q$, we have

$\mathbb{I}_A - \mathbb{I}_B = \mathbb{I}_{\{ g_*(x) \neq g(x) \}} \mathbb{I}_{\{ Y = g_*(x) \}}$

$- \mathbb{I}_{\{ g(x) \neq g_*(x) \}} \mathbb{I}_{\{ Y = g(x) \}}$

$= \mathbb{I}_{\{ g_*(x) \neq g(x) \}} \left[\mathbb{I}_{\{ Y = g_*(x) \}} - \mathbb{I}_{\{ Y = g(x) \}} \right]$

$$\therefore \mathcal{E}(g) = E_{(X,Y)\sim D}\left[\mathbb{1}_A - \mathbb{1}_B\right]$$

$$= E_{(X,Y)\sim D}\Big[\overbrace{\mathbb{1}_{\{g(x)\neq g_*(x)\}}}^{h(x)} \cdot \left(\mathbb{1}_{\{Y=g_*(x)\}} - \mathbb{1}_{\{Y=g(x)\}}\right)\Big]$$

$$= E_{X\sim\pi} E_{Y\sim P(\cdot|x)}\left[\underbrace{h(x)}_{\text{constant w.r.t. } Y}\left(\mathbb{1}_{\{Y=g_*(x)\}} - \mathbb{1}_{\{Y=g(x)\}}\right)\right]$$

$$= E_{X\sim\pi}\, h(x)\, E_{Y\sim P(\cdot|x)}\left[\mathbb{1}_{\{Y=g_*(x)\}} - \mathbb{1}_{\{Y=g(x)\}}\right]$$

conditional expectation of indicator = conditional probability

$$= E_{X\sim\pi}\, h(x)\left[\overbrace{P(Y=g_*(x)\mid x)}^{\downarrow} - \overbrace{P(Y=g(x)\mid x)}^{\downarrow}\right]$$

$$= E_{X\sim\pi}\left[\mathbb{1}_{\{g(x)\neq g_*(x)\}}\left[P(Y=g_*(x)\mid x) - P(Y=g(x)\mid x)\right]\right]$$

$$= E_{X\sim\pi}\left[\mathbb{1}_{\{g(x)=1,\, g_*(x)=-1\}\cup\{g(x)=-1,\, g_*(x)=+1\}}\left[P(Y=g_*(x)\mid x) - P(Y=g(x)\mid x)\right]\right]$$

Since $\mathbb{1}_{A\cup B} = \mathbb{1}_A + \mathbb{1}_B$ for $A\cap B = \emptyset$, we get

$$\mathcal{E}(g) = E_{X\sim\pi}\left[\left[\mathbb{1}_{\{g(x)=1,\, g_*(x)=-1\}} + \mathbb{1}_{\{g(x)=-1,\, g_*(x)=1\}}\right] \times \left[P(Y=g_*(x)\mid x) - P(Y=g(x)\mid x)\right]\right]$$

$$\Rightarrow \mathcal{E}(g) = E_{x\sim\pi}\Big[\, 1_{\overbrace{\{g(x)=1,\, g_*(x)=-1\}}^{M}} \cdot \big(P(Y=g_*(x)\mid x) - P(Y=g(x)\mid x)\big)\Big]$$

$$+\, E_{x\sim\pi}\Big[\, 1_{\overbrace{\{g(x)=-1,\, g_*(x)=+1\}}^{N}} \cdot \big(P(Y=g_*(x)\mid x) - P(Y=g(x)\mid x)\big)\Big]$$

$$\Rightarrow \mathcal{E}(g) = E_{x\sim\pi}\, 1_M \big(P(Y=-1\mid x) - P(Y=+1\mid x)\big)$$

$$+\, E_{x\sim\pi}\, 1_N \big(P(Y=+1\mid x) - P(Y=-1\mid x)\big)$$

Recall that $P(Y=1\mid X) = p_x$.

And $P(Y=-1\mid X) = 1-p_x$

$$\therefore \mathcal{E}(g) = E_{x\sim\pi}\big[1_M\,(1-p_x-p_x)\,\big]$$

$$+\, E_{x\sim\pi}\big[\,1_N\,(p_x-(1-p_x)\,\big]$$

$$= E_{x\sim\pi}\big[\,1_M\,(1-2p_x)\big]$$

$$+\, E_{x\sim\pi}\big[\,1_N\,(2p_x-1)\,\big]$$

$$= E_{x\sim\pi}\big[\,(1_N-1_M)\,(2p_x-1)\,\big]$$

$$\Rightarrow \mathcal{E}(g) = E_{x \sim \pi}\left[(1_N - 1_M)(2p_x - 1) \right]$$

Recall that $\eta(x) = E[Y \mid X]$

$$= \sum_{y \in \{+1,-1\}} y \, P(Y = y \mid X)$$

$$= +1 \cdot P(Y=1 \mid x) - 1 \, P(Y=-1 \mid x)$$

$$= p_x - (1 - p_x) = 2p_x - 1$$

$$\therefore \mathcal{E}(g) = E_{x \sim \pi}\left[(1_N - 1_M)\, \eta(x) \right]$$

Also recall that $M = \{ g(x) = 1, g_*(x) = -1 \}$

$$N = \{ g(x) = -1, g_*(x) = +1 \}$$

$$\therefore M = \{ g(x) \neq g_*(x) \} \cap \{ g_*(x) = -1 \}$$

$$N = \{ g(x) \neq g_*(x) \} \cap \{ g_*(x) = +1 \}$$

But $g_*(x) = 1 \iff p_x \geq 1/2$

& $g_*(x) = -1 \iff p_x < 1/2$

$$\therefore M = \{ g(x) \neq g_*(x) \} \cap \{ p_x < 1/2 \}$$

$$N = \{ g(x) \neq g_*(x) \} \cap \{ p_x \geq 1/2 \}$$

$p_x < 1/2$ | $p_x \geq 1/2$

$\Rightarrow 2p_x - 1 < 0$ | $\Rightarrow 2p_x - 1 \geq 0$

$\Rightarrow \eta(x) < 0$ | $\Rightarrow \eta(x) \geq 0$.

So $M = \{ g(x) \neq g_*(x) \} \cap \{ \eta(x) < 0 \}$

Ω

$N = \{ g(x) \neq g_*(x) \} \cap \{ \eta(x) \geq 0 \}$

$\therefore 1_M = 1_\Omega 1_{\{\eta(x) < 0\}}$

$1_N = 1_\Omega 1_{\{\eta(x) \geq 0\}}$

$\therefore \mathcal{E}(g) = E_{x \sim \pi}\left[(1_N - 1_M)\, \eta(x) \right]$

$= E_{x \sim \pi}\left[\left(1_N \cdot 1_{\{\eta(x) \geq 0\}} - 1_M \cdot 1_{\{\eta(x) < 0\}} \right) \eta(x) \right]$

$= E_{x \sim \pi}\left[1_\Omega 1_{\{\eta(x) \geq 0\}}\, \eta(x) \right.$

$\left. - 1_\Omega 1_{\{\eta(x) < 0\}}\, \eta(x) \right]$

$= E_{x \sim \pi}\left[1_\Omega \left\{ 1_{\{\eta(x) \geq 0\}}\, \eta(x) \right.\right.$

$\left.\left. + 1_{\{\eta(x) < 0\}} (-\eta(x)) \right\} \right]$

$$\Rightarrow \mathcal{E}(g) = E_{x\sim\pi}\left[1_{\mathcal{Q}} \cdot |\eta(x)| \right]$$

$$= E_{x\sim\pi}\left[1\{g(x) \neq g_*(x)\} \cdot |\eta(x)| \right]$$

$$= \int_X 1\{g(x) \neq g_*(x)\} \cdot |\eta(x)| \, \pi(dx)$$

$$= \int_{\{x : g(x) \neq g_*(x)\}} |\eta(x)| \, \pi(dx)$$

③ $g(x) = \text{sign}(\hat{\eta}(x))$ for some $\hat{\eta} : S \to \mathbb{R}$

$$\mathcal{E}(g) = \int_{\{x : g(x) \neq g_*(x)\}} |\eta(x)| \, \pi(dx)$$

Let $T := \{x : g(x) \neq g_*(x)\}$, it means

$g(x) = +1, \; g_*(x) = -1$ or $g(x) = -1, g_*(x) = +1$

So, $g(x) \cdot g_*(x) < 0 \Rightarrow \text{sign}(\hat{\eta}(x)) \text{sign}(\eta(x)) < 0$

$\Rightarrow \text{sign}(\hat{\eta}(x) \cdot \eta(x)) < 0$

$\Rightarrow \hat{\eta}(x) \cdot \eta(x) < 0$

If $\hat{\eta}(x)\,\eta(x) < 0$

$$\Rightarrow |\hat{\eta}(x) - \eta(x)| = |\hat{\eta}(x)| + |\eta(x)|$$

$$\Rightarrow |\eta(x)| \leq |\hat{\eta}(x) + \eta(x)| = |\hat{\eta}(x) - \eta(x)|$$

$$\Rightarrow |\eta(x)| \leq |\hat{\eta}(x) - \eta(x)|$$

$$\Rightarrow \int_T |\eta(x)|\pi(dx) \leq \int_T |\hat{\eta}(x) - \eta(x)|\,\pi(dx)$$

$$\Rightarrow \int_T |\eta(x)|\,\pi(dx) \underset{\Delta}{\leq} \int_T |\hat{\eta}(x) - \eta(x)|\,\pi(dx)$$

$$+ \int_{S\setminus T} |\hat{\eta}(x) - \eta(x)|\,\pi(dx)$$

where $S\setminus T = \{x : g(x) = g_*(x)\}$ (could be empty) Δ

Note: $g(x) = g_*(x)$ says $\hat{\eta}(x)$ & $\eta(x)$ have the same sign, but not that they are equal.

$$\therefore \int_T |\eta(x)|\,\pi(dx) \leq \int_S |\hat{\eta}(x) - \eta(x)|\,\pi(dx)$$

Now, for any function f, $\|f\|_{L^1(\pi)} \leq \|f\|_{L^2(\pi)}$

Proof: Using Cauchy-Schwarz

$$\|f\|_{L^1(\pi)} = \int_S |f| \underbrace{d\pi}_{\pi(dx)} = \int_S |f| \cdot 1 \, d\pi = \langle |f|, 1 \rangle_\pi$$

$$\Rightarrow \langle |f|, 1 \rangle_\pi \leq |f|_\pi \, |1|_\pi$$

$$\Rightarrow \langle |f|, 1 \rangle_\pi^2 \leq \underbrace{|f|_\pi^2}_{\downarrow} \underbrace{|1|_\pi^2}_{\rightarrow}$$

$$= \langle |f|, |f| \rangle_\pi \langle 1, 1 \rangle_\pi$$

$$= \int_S |f|^2 d\pi \cdot \underbrace{\int_S 1^2 \, d\pi}_{\text{Integration of the probability measure over all space} = 1}$$

$$\Rightarrow \|f\|_{L^1(\pi)}^2 \leq \underbrace{\int_S |f|^2 d\pi}_{\|f\|_{L^2(\pi)}}$$

$$\Rightarrow \|f\|_{L^1(\pi)} \leq \|f\|_{L^2(\pi)}^{1/2}$$

Setting $f(x) = \hat{\pi}(x) - \pi(x)$,

$$\int_S |\hat{\pi}(x) - \pi(x)|\, d\Pi \leq \left(\int_S |\hat{\pi}(x) - \pi(x)|^2 \, d\Pi \right)^{1/2}$$

From earlier, we had

$$\underbrace{\int_{T=\{\pi : g(\pi) \neq g_*(\pi)\}} |\pi(x)|\, d\Pi}_{\mathcal{E}(g)} \leq \int_S |\hat{\pi}(x) - \pi(x)|\, d\Pi$$

$$\leq \left(\int_S |\hat{\pi}(x) - \pi(x)|^2 \, d\Pi \right)^2$$

PROBLEM : 2

$$g_t(x) = \begin{cases} +1 & x \geq t \\ -1 & x < t \end{cases} \qquad t \in \mathbb{R}$$

$g_t(x) = Sgn(x - t)$ where

$$Sgn(z) = \begin{cases} +1, & z \geq 0 \\ -1, & z < 0 \end{cases}$$

Also $Y_i = g_{t_*}(X_i)$

L_S = Emperical loss.

$$L_S = \sum_{i=1}^{n} \ell(g_t(X_i), Y_i) = \sum_{i=1}^{n} \mathbb{1}\{g_t(x_i) \neq Y_i\}$$

$$= \sum_{i=1}^{n} \mathbb{1}\{g_t(x_i) \neq g_{t_*}(x_i)\}$$

$$= \sum_{i=1}^{n} \frac{1 - \mathbb{1}\{g_t(x_i)\}\mathbb{1}\{g_{t_*}(x_i)\}}{2}$$

$$= \sum_{i=1}^{n} \frac{1 - Sgn(x_i - t)\, Sgn(x_i - t_*)}{2}$$

$$= \sum_{i=1}^{n} \frac{1 - \mathrm{Sgn}(x_i - t)\,\mathrm{Sgn}(x_i - t_*)}{2}$$

L_S is minimized when this term is maximized

$$\max_t \text{term} = \sum_{i=1}^{n} \mathrm{Sgn}(x_i - t)\,\mathrm{Sgn}(x_i - t_*)$$

So we want as many x_i to lie on the same side of both t & t_*. But t_* is fixed. Hence the maximum is achieved for $t \in (x_k, x_{k+1}]$ where $x_k < t_* \leq x_{k+1}$

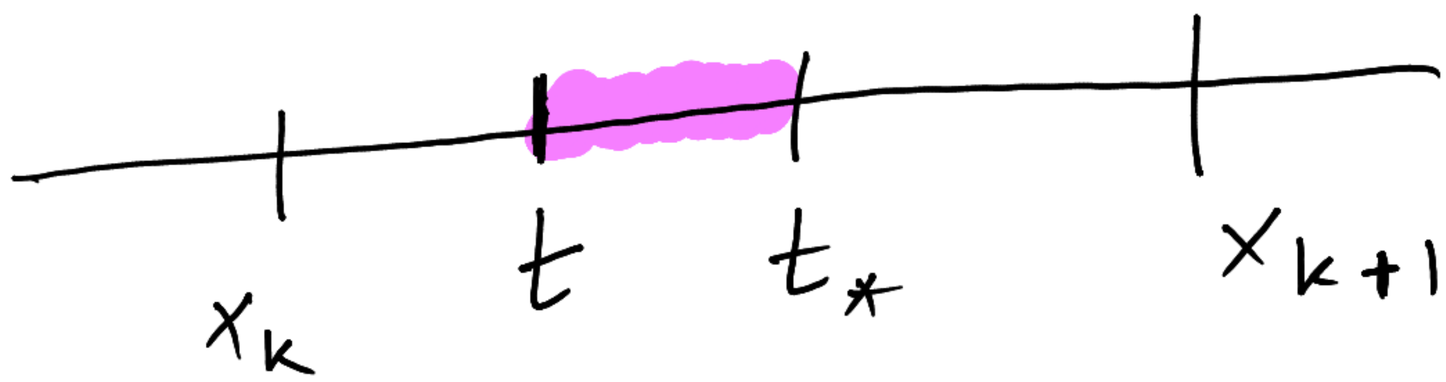
Hence the ERM returns a $g_t(x)$ where. $t \in (x_k, x_{k+1}]$.

In other words:-

$\forall\, t \in [a, b]$, $a = x_k$, $b = x_{k+1}$, $g_t(x)$ is an ERM



② Let $\epsilon > 0$. We want to estimate $Pr(L(\hat{g}) \geq \epsilon)$ where $\hat{g}$ is any ERM solution.



Any time a data point falls in between t & t_*, the loss is 1.

$$L(\hat{g}) = \mathop{\mathbb{E}}_{x \sim D}\left[\ell\left(g_t(x), g_{t_*}(x) \right) \right]$$

$$= \mathop{\mathbb{E}}_{x \sim D}\left[\mathbb{1}\{ g_t(x) \neq g_{t_*}(x) \} \right]$$

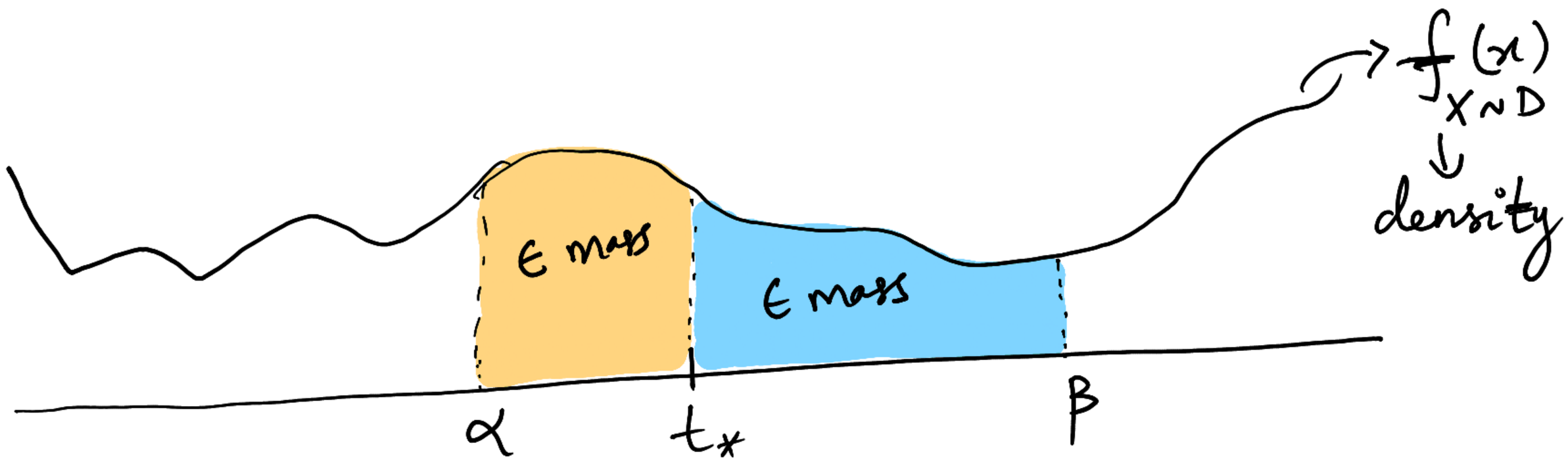
$$= \mathop{\mathbb{E}}_{x \sim D}\left[\mathbb{1}\{ \mathrm{Sgn}(x-t) \neq \mathrm{Sgn}(x-t_*) \} \right]$$

$$= \mathop{\mathbb{E}}_{x \sim D}\left[\mathbb{1}\{ x \in [\min(t, t_*), \max(t, t_*))\} \right]$$

$$= \mathop{P}_{x \sim D}\left(x \in [\min(t, t_*), \max(t, t_*)) \right)$$

= Chance that $x \sim D$ falls between t & t_*.

Now, let us vizualize the distribution $X \sim D$.

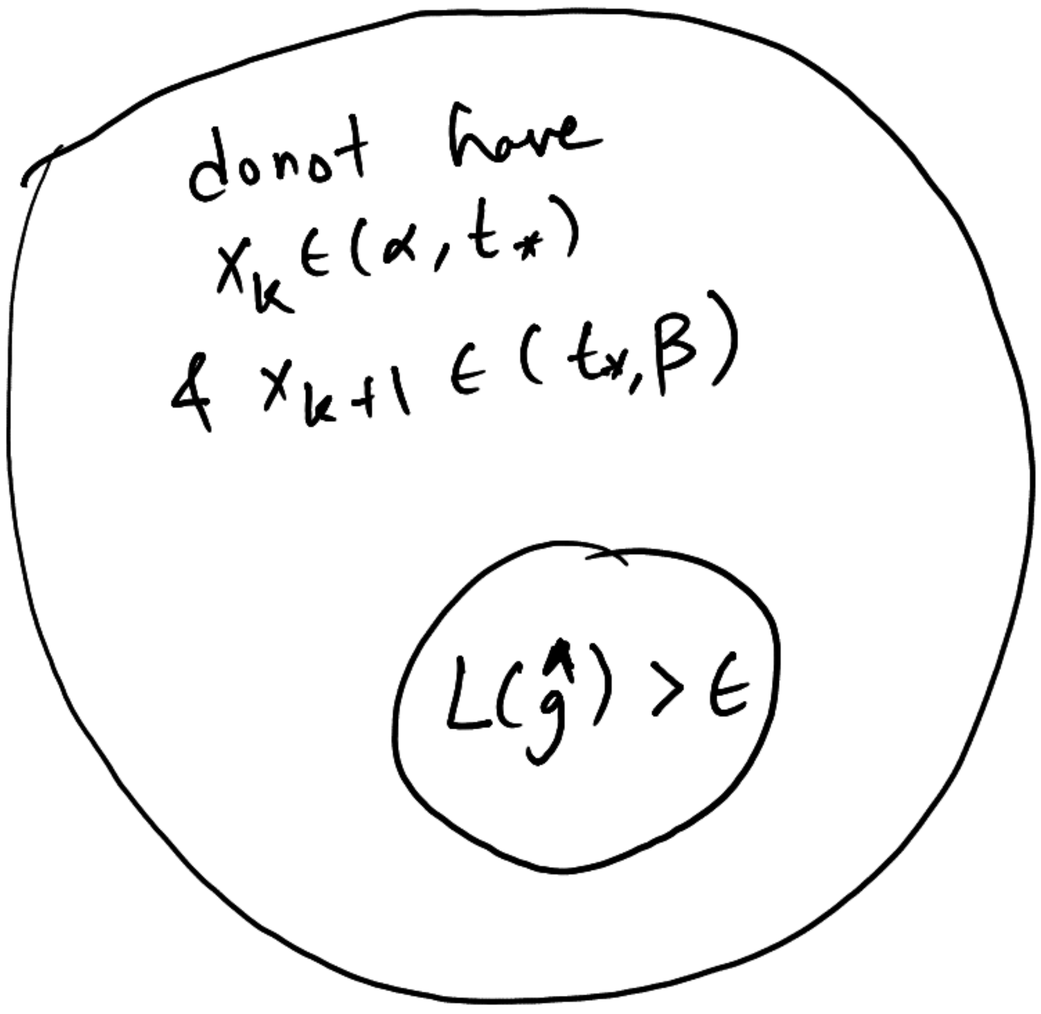


If $\hat{g}_t$ is an ERM, as long as $\alpha < t < \beta$, the loss $L(g_t)$ is going to be less than ϵ.

We need at least 2 samples X_k & X_{k+1} such that $X_k \in (\alpha, t_*)$ & $X_{k+1} \in (t_*, \beta)$ to guarantee $L(\hat{g}) < \epsilon$.

If we do not have two such samples, then $L(\hat{g})$ may be greater than ϵ.

The event $\{ L(\hat{g}) > \epsilon \}$ can happen only if we do not have 2 such samples.



$$\therefore P(L(\hat{g}) > \epsilon)$$

$$\leq P(\text{no two such } x_i \sim D \text{ samples})$$

$$P_{x_i \sim D}(\text{no two such sample})$$

$$= P_{x_i \sim D}(\text{either } \nexists\, x_i \in (\alpha, t_*) \quad \nexists\, x_i \in (t_*, \beta))$$

$$= P_{x_i \sim D}(\nexists x_i \in (\alpha, t_*))$$

$$\sim + P_{x_i \sim D}(\exists x_i \in (t_*, \beta))$$

$$= P_{x_i \sim D}(x_1 \notin (\alpha, t_*) \,\&\, \ldots \,\&\, x_n \notin (\alpha, t_*))$$

$$+ P_{x_i \sim D}(x_1 \notin (t_*, \beta) \,\&\, \ldots \,\&\, x_n \notin (\alpha, t_*))$$

independence

$$= \prod_{i=1}^{n} P_{x_i \sim D}(x_i \notin (\alpha, t_*)) + \prod_{i=1}^{n} P_{x_i \sim D}(x_i \notin (t_*, \beta))$$

identical distribution

$$= \left[P_{x_1 \sim D}(x_1 \notin (\alpha, t_*))\right]^n + \left[P_{x_1 \sim D}(x_1 \notin (t_*, \beta))\right]^n$$

$$\therefore \quad P(L(\hat{g}) > \epsilon) \leq (1-\epsilon)^n + (1-\epsilon)^n$$

$$\leq 2(1-\epsilon)^n.$$

$\forall x \in \mathbb{R}$
$1+x \leq \exp(x)$

$$\leq 2\exp(-n\epsilon)$$

(3) $|G| < \infty$, $G = \{g_1 \ldots g_n\}$

Description of the ERM in Agnostic Case :-

Here, the ERM is of the form $\hat{g}_t(x)$ where t is such that $\sum_{i=1}^{n} \ell(\hat{g}_t(x_i), y_i)$ is minimized. This can be done by checking $L_S(\hat{g}_t)$ for $t = x_{(1)} - 1, \frac{x_{(1)} + x_{(2)}}{2}, \ldots \ldots \frac{x_{(n-1)} + x_{(n)}}{2}, x_{(n)} + 1$ and picking t that minimizes $L_S(\hat{g}_t)$.

$x_{(1)} \ldots \ldots x_{(n)}$ are the samples sorted.

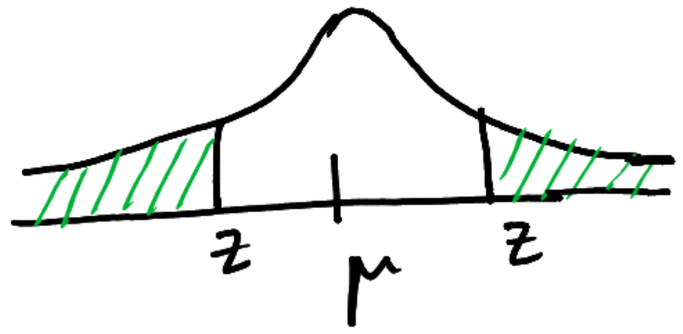
Let us recall Hoeffding's Inequality :-

If Z is σ- Sub-Gaussian with mean μ,

then $P(|Z-\mu| > t) \leq 2\exp(-t^2/2\sigma^2)$

Also, $P(Z-\mu > t) \leq \exp(-t^2/2\sigma^2)$

Also, $P(\mu - Z > t) \leq \exp(-t^2/2\sigma^2)$



Now, for a fixed $g \in G$,

$$L_S(g) = \frac{1}{n}\sum_{i=1}^{n} \ell(f(x_i), Y_i)$$

$$L(g) = \mathop{\mathbb{E}}_{X \sim D}[\ell(f(x), Y)]$$

Now, $\mathop{\mathbb{E}}_{S \sim D^n} L_S(g) = \mathop{\mathbb{E}}_{X \sim D}[\ell(f(x), Y)] = L(g)$

Also, $\ell_i = \ell(f(x_i), Y_i) \in \{0, 1\}$

Lemma: ℓ_i are $\frac{1}{2}$- Sub-Gaussian.

Let us recall Sub-Gaussian Calculus :-

i) If Z is σ-Sub Gaussian, cZ is $|c|\sigma$ Sub-Gaussian.

ii) If Z_i are σ_i- Sub Gaussian, $\sum Z_i$ are $\sqrt{\sum \sigma_i^2}$ Sub-Gaussian

$\frac{1}{n}\ell_i$ are $\frac{1}{n}$ Sub-Gaussian & hence $\sum_{i=1}^{n} \frac{1}{n}\ell_i = L_S(g)$

is $\frac{1}{2\sqrt{n}}$ Sub-Gaussian.

Hence $L_S(g)$ is $\frac{1}{2\sqrt{n}}$ Sub-Gaussian.

$$\Rightarrow P\left(\left| L_S(g) - \mathop{\mathbb{E}}_{S\sim D^n} L_S(g) \right| > \epsilon \right)$$

$$\Rightarrow P\left(|L_S(g) - L(g)| \geq t \right) \leq 2\exp\left(-t^2 / 2\left(\tfrac{1}{2\sqrt{n}}\right)^2\right)$$

$$= 2\exp(-2nt^2)$$

Now, this probability is for a single $g \in G$.

We are asked about $P(L(\hat{g}) - L(\bar{g}) \geq \epsilon)$

where $\bar{g}$ is the best in G w.r.t L_S.

Let the event E be $\sup_{g\in G} |L(g) - L_S(g)| < t$.

Under the event E, $L(\hat{g}) - L_S(\hat{g}) < t$

$$\Rightarrow L(\hat{g}) - L_S(\bar{g}) < t \rightarrow ①$$

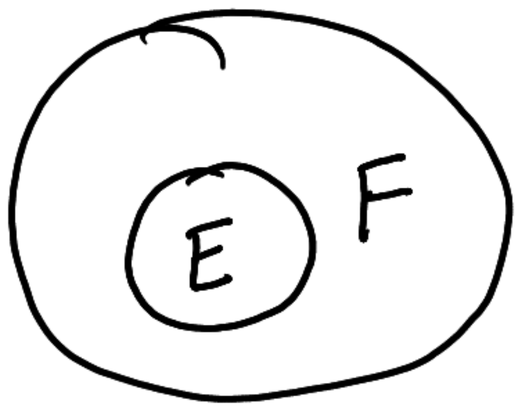
Also under E, $L_S(\bar{g}) - L(\bar{g}) < t \rightarrow ②$

$① + ② \Rightarrow L(\hat{g}) - L(\bar{g}) < 2t$

Under the event $E = \left\{ \sup_{g \in G} |L(g) - L_S(g)| < t \right\}$

implies $F = \{ L(\hat{g}) - L(\bar{g}) < 2t \}$

$E \Rightarrow F$



$\Rightarrow F \supseteq E$

$\Rightarrow F^c \subseteq E^c$

$\Rightarrow P(F^c) \leq P(E^c)$

$F^c = \{ L(\hat{g}) - L(\bar{g}) \geq 2t \}$

$E^c = \left\{ \sup_{g \in G} |L(g) - L_S(g)| \geq t \right\}$

$= \{ \exists g \in G, |L(g) - L_S(g)| \geq t \}$

$= \bigcup_{i=1} \{ |L(g_i) - L_S(g_i)| \geq t \}$

$\Rightarrow P(E^c) = P\left(\bigcup_{i=1} \{ |L(g_i) - L_S(g_i)| \geq t \} \right)$

$$\Rightarrow P(E^c) \leq \sum_{i=1}^{m} P\left(\{|L(g_i) - L_S(g_i)| \geq t\}\right)$$

Union bound

$\frac{1}{\sqrt{n}}$-Sub-Gaussian

$$\Rightarrow P(E^c) \leq \sum_{i=1}^{m} 2\exp(-2nt^2)$$

Hoeffding Inequality

$$\Rightarrow P(E^c) \leq 2m \exp(-2nt^2)$$

$$\Rightarrow P(F^c) \leq P(E^c) \leq 2m \exp(-2nt^2)$$

$$\Rightarrow P(L(\hat{g}) - L_S(\bar{g}) \geq 2t) \leq 2m \exp(-2nt^2)$$

Setting $2t = \epsilon$,

$$P(L(\hat{g}) - L_S(\bar{g}) \geq \epsilon) \leq 2m \exp\left(-\frac{n\epsilon^2}{2}\right)$$

Q3 $Z_i = I\{X_i \leq x\}$ is a Bernoulli random variable.

Also Z_i are i.i.d.

Hence by Weak Law of Large Numbers,

$$F_n(x) \longrightarrow F(x).$$

Bounding the expression

$$\left| p(x) - \frac{F(x+h) - F(x-h)}{2h} \right| \leq ?$$

where $p(x) = F(x)$

Note: $F(x+h) = F(x) + hF'(x) + \frac{h^2}{2}F''(x) + \frac{h^3}{6}F'''(\alpha)$

for some $\alpha \in (x, x+h)$

$F(x-h) = F(x) - hF'(x) + \frac{h^2}{2}F''(x) - \frac{h^3}{6}F'''(\beta)$

for some $\beta \in (x-h, x)$.

$$F(x+h) - F(x-h) = 2hF'(x) + \frac{h^3}{6}\left(F'''(\alpha) + F'''(\beta)\right)$$

$$\therefore \left| p(x) - \frac{F(x+h) - F(x-h)}{2h} \right|$$

$$= \left| \underbrace{p(x)}_{F'(x)} - F'(x) - \frac{h^3}{12}\left(F'''(\alpha) + F'''(\beta)\right) \right|$$

$$= \left| \frac{h^2}{12}\left(F'''(\alpha) + F'''(\beta)\right) \right|$$

$$\leq \frac{h^2}{12}\left(\left|F'''(\alpha)\right| + \left|F'''(\beta)\right|\right)$$

If $F'''(x)$ is bounded by M, then

$$\left| p(x) - \frac{F(x+h) - F(x-h)}{2h} \right| \leq \frac{h^2 M}{12}.$$

We are asked to show that replacing F by Fn in the central difference approximation of p(x) yeilds the kernel density estimator for a specific kernel function.

$$\hat{p}(x) = \frac{F_n(x+h) - F_n(x-h)}{2h}$$

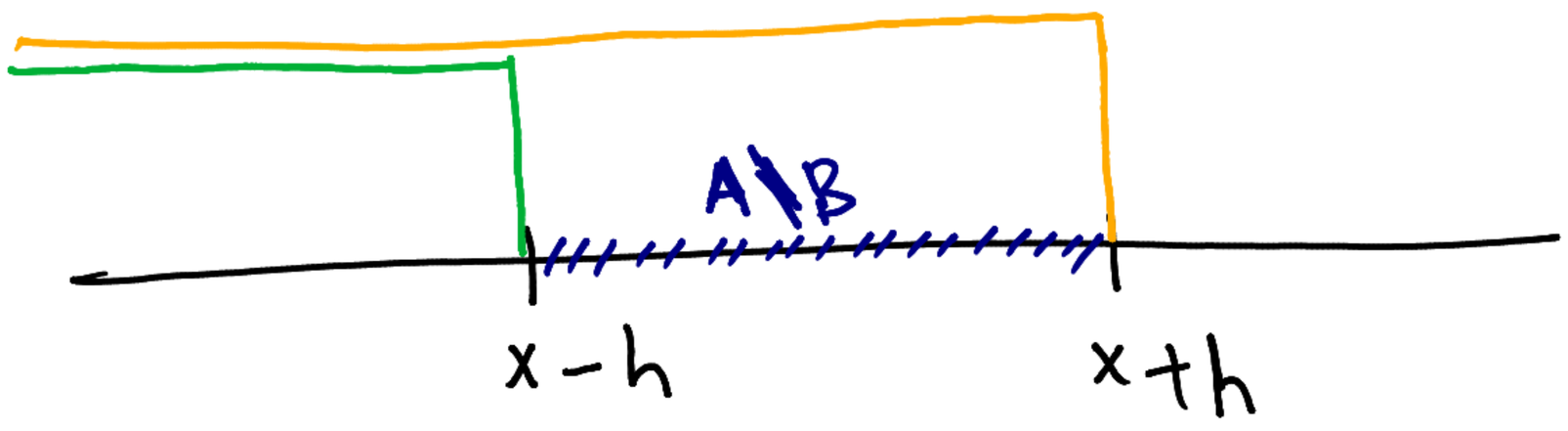
$$= \frac{1}{2h}\left[\frac{1}{n}\sum_{j=1}^{n} \mathbb{1}\{x_j \le x+h\} - \frac{1}{n}\mathbb{1}\{x_j \le x-h\}\right]$$

$$= \frac{1}{2nh}\left[\sum_{j=1}^{n} \mathbb{1}\{x_j \le x+h\} - \mathbb{1}\{x_j \le x-h\}\right]$$

If $A \supseteq B$, then $\mathbb{1}_A - \mathbb{1}_B = \mathbb{1}_{\{A \setminus B\}}$

$$A = \{x_j \le x+h\}, \quad B = \{x_j \le x-h\}$$

$$A \setminus B = \{x-h \le x_j \le x+h\}$$



$$\therefore \hat{p}(x) = \frac{1}{2nh} \sum_{j=1}^{n} \mathbb{1}\{x-h \leq X_j \leq x+h\}$$

$$= \frac{1}{nh} \sum_{j=1}^{n} K\left(\frac{x - X_j}{h}\right)$$

where the kernel is

$$K(u) = \frac{1}{2} \mathbb{I}\{|u| \leq 1\}$$

and is called the uniform kernel.

(b) $K(x) = I\left\{|x| \leq 1/2\right\}$, $x \in \mathbb{R}$.

If $p_X(x)$ wasn't known, the estimator was

$$\hat{m}(x) = \frac{\sum_{i=1}^{n} K\left(\frac{x - X_i}{h/2}\right) Y_i}{\sum_{i=1}^{n} K\left(\frac{x - X_i}{h/2}\right)}$$

$$= \frac{\sum_{i=1}^{n} \frac{1}{2} \mathbb{I}\left\{\left|\frac{x - X_i}{h/2}\right| \leq 1\right\} Y_i}{\sum_{i=1}^{n} \frac{1}{2} I\left\{\left|\frac{x - X_i}{h/2}\right| \leq 1\right\}}$$

$$\Rightarrow \hat{m}(x) = \frac{\sum_{i=1}^{n} \mathbb{I}\{X_i \in [x-h/2,\ x+\frac{h}{2}]\, Y_i}{\sum_{i=1}^{n} \mathbb{I}\{X_i \in [x-\frac{h}{2},\ x+\frac{h}{2}]\}}$$

$$\mathbb{I}\{x_i \in [x-h/2,\ x+\tfrac{h}{2}]\} \cong p_x(x)\cdot h$$

$$\therefore \hat{m}(x) = \frac{\sum_{i=1}^{n} \mathbb{I}\{X_i \in [x-\frac{h}{2},\ x+\frac{h}{2}]\, Y_i}{\sum_{i=1}^{n} p_x(x)\cdot h}$$

$$= \frac{1}{nh\,p_x(x)} \sum_{i=1}^{n} \mathbb{I}\{X_i \in [x-h/2,\ x+\tfrac{h}{2}]\}$$